

Homework set 3

Due March 19, 2026, at 14:00

via <https://elo.mastermath.nl/mod/assign/view.php?id=38195>.

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Definitions

Def 1 (*Rademacher Complexity*): The Rademacher Complexity of loss function ℓ , hypothesis class $\mathcal{H}$ and sample $S = ((\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_m, Y_m))$ is defined as

$$\mathcal{R}(\ell, H, S) = \frac{1}{m} \mathbb{E} \left[\sup_{h \in H} \sum_{i=1}^m \sigma_i \ell(h, \mathbf{X}_i, Y_i) \right] \quad (1)$$

where $\sigma_i \in \{\pm 1\}$ are i.i.d. Rademacher random variables.

Def 2 (*L-Lipschitz*): A function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is *L-Lipschitz* when $|\varphi(\alpha) - \varphi(\beta)| \leq L|\alpha - \beta|$.

Exercise 1 [2pt]

Exercise description here:

- (a)
- (b)
- (c)

Answer to Exercise

- (a)
- (b)
- (c)

Exercise 2 [2pt]

Answer to Exercise 2